

Cevahir Koprulu

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RESEARCH INTERESTS

Keywords: Reinforcement learning (RL), Generalization, and Curriculum Learning

My research focuses on sample-efficient RL: what tasks an agent trains on and what signal it learns from. I designed automated curriculum generation methods for heavy-tailed task distributions, task specifications, cost constraints, and the scale of batched autonomous driving simulators. I also developed methods for reward synthesis from task-agnostic experience with a few demonstrations and from physics-informed learned dynamics models with distance-aware uncertainty estimates. These works span offline RL and learning from demonstrations. Ongoing work extends curriculum generation to composing traffic behaviors for self-play RL and to unlearning in large language models.

EDUCATION

University of Texas at Austin, Austin, TX, USA
Electrical and Computer Engineering

M.S. 2023, PhD 2027
GPA: 3.94/4

Bilkent University, Ankara, Turkey
Electrical and Electronics Engineering

B.Sc. June 2021
GPA: 3.73/4

Honours/Awards: (i) Bilkent University Comprehensive Scholarship (ii) Scholarship of Turkish Prime Ministry (iii) Bilkent University EEE Department High Honours (iv) Ranked 24th among 2 million students in university entrance exam (v) FIRST Robotics Competition 2015: Rookie All-Star

WORK EXPERIENCE

Waymo

Software Engineering Intern (all work is under NDA) || Python, SQL, JAX

Mountain View, CA
June 2026 - August 2026

- Researched **learning from unsupervised driving data within Waymo's data flywheel** for onboard planner models, with a focus on undesirable driving behaviors. Surveyed state-of-the-art methods, implemented and compared a selection within the training infrastructure, and proposed an algorithm and data roadmap for future deployment.

Bosch Center for Artificial Intelligence

Research Intern (all work is under NDA) || Python, PyTorch

Sunnyvale, CA
May 2025 - August 2025

- Developed **CL4AD**, the *first* integration of curriculum learning into GPUDrive, a batched autonomous driving simulator, reducing wall clock time to achieve 99% success rate by 77%, more than a billion steps earlier than domain randomization, i.e., the default uniform scenario sampling method.

Honda Research Institute - US

Research Intern (all work is under NDA) || Python, PyTorch, RAY

Ann Arbor, MI
May 2024 - August 2024

- Developed an **action advising framework**, GEN2SPEC, that distills knowledge of generalist agents to train specialist agents in a continual learning setting. This framework accelerates the training of an RL agent via action advice from a transformer-based meta-RL agent that can adapt to unseen tasks.

Eatron Technologies

Engineering Intern (all work is under NDA) || Python, C++, PyTorch, ROS

Istanbul, Turkey
June 2020 - June 2021

- Designed **graph convolutional** and **self-attention**-based neural network architectures to extract spatio-temporal features of a traffic scene for trajectory prediction in a Level-2+ ADAS powered vehicle.

ROKETSAN (in collaboration with Bilkent University)

Industrial Design Project (all work is under NDA) || Python, C++, PyTorch, ROS

Ankara, Turkey
Sept 2019 - June 2020

- Developed a mobile robot that follows a human leader, combining **YOLOv3 for object detection** and **artificial potential field method for path planning** in a mapped area with unknown dynamic obstacles.

RESEARCH WORK

[Scaling Curriculum Learning For Autonomous Driving](#)

ArXiv, 2026

Cevahir Koprulu, David Paz, Feng Tao, Yuliang Guo, Xinyu Huang, Ufuk Topcu, Liu Ren

[Neural Stochastic Differential Equations for Uncertainty-Aware, Offline Reinforcement Learning](#)

ICLR, 2025

Cevahir Koprulu, Franck Djeumou, Ufuk Topcu

[Safety Prioritizing Curricula for Constrained Reinforcement Learning](#)

ICLR, 2025

Cevahir Koprulu, Thiago D. Simão, Nils Jansen, Ufuk Topcu

Dense Dynamics-Aware Reward Synthesis: Integrating Prior Experience with Demonstrations Cevahir Koprulu , Po-han Li, Tianyu Qiu, Ruihan Zhao, Tyler Westenbroek, David Fridovich-Keil, Sandeep P. Chinchali, Ufuk Topcu	L4DC, 2025
Joint Learning of Reward Machines and Policies in Environments with Partially Known Semantics Christos Verginis, Cevahir Koprulu , Sandeep Chinchali, Ufuk Topcu	Artificial Intelligence, 2024
Risk-Aware Curriculum Generation for Heavy-tailed Task Distributions Cevahir Koprulu , Thiago D. Simão, Nils Jansen, Ufuk Topcu	UAI, 2023
Reward-Machine-Guided, Self-Paced Reinforcement Learning Cevahir Koprulu , Ufuk Topcu	UAI, 2023
Act to Reason: A Dynamic Game Theoretical Driving Model for Highway Merging Applications Cevahir Koprulu , Yildiray Yildiz	CCTA, 2021

TECHNICAL SKILLS

- Programming languages: (Expert) Python, (Proficient) C++, Julia
- Frameworks: PyTorch, JAX, TensorFlow, ROS, RAY, verl, vLLM, Hugging Face Transformers

RELEVANT COURSEWORK

- **UT Austin:** Online Learning, Causality and Reinforcement Learning, Statistical Machine Learning, Learning-based Optimal Control, Game-Theoretic Modeling of Multi-Agent Systems, Program Synthesis, Cyber-Physical Systems, Reinforcement Learning, Convex Optimization, Probability and Statistics, and Complex Networks in the Real World.
- **École polytechnique fédérale de Lausanne** (Exchange - Spring 2019): Image Analysis and Pattern Recognition, Convex Optimization, Deep Learning, and Biological Modelling of Neural Networks.
- **Bilkent University:** Statistical Learning and Data Analytics, Robust Feedback Theory, Introduction to Financial Mathematics, Neural Networks, and Game Theory.

RESEARCH AND LEADERSHIP EXPERIENCES

Reinforcement Learning Reading Group <i>Organizer</i>	Austin, TX Aug 2025 - Ongoing
• RLRG at the CS Department of UT Austin is a student-run group that discusses research papers on RL.	
Center for Autonomy <i>Ph.D. Student</i>	Austin, TX Aug 2021 - Ongoing
Systems Laboratory <i>Undergraduate Researcher</i>	Ankara, Turkey Sept 2019 - July 2021
• Developed human driver models from real-traffic data that can change its reasoning level dynamically by combining level-k game theory and reinforcement learning.	
IEEE Robotics and Automation Society at Bilkent University <i>Chairman</i>	Ankara, Turkey May 2017 - June 2018
• Organized “Mühendis Kafası” in cooperation with Technology Development Foundation of Turkey: Series of sessions on Computer Vision and Deep Learning. • Gave lectures on robotics, control techniques, and related micro-controller programming: EE-101: Introduction to Robotics with Arduino.	

LANGUAGE SKILLS

- Turkish: Native proficiency
- English: TOEFL 110/120 (Fall 2020)
- French: DELF B1 (Spring 2015)

RECREATIONAL INTERESTS

I enjoy climbing, cycling, watching/playing soccer (football :)), and learning about history and psychology.